SciFact controlled format benchmark

Combination varenicline and bupropion SR for tobacco-dependence treatment in cigarette smokers: a randomized trial.

SCIFACT-EVIDENCE-11614737

Evidence abstract

IMPORTANCE Combining pharmacotherapies for tobacco-dependence treatment may increase smoking abstinence.

OBJECTIVE To determine efficacy and safety of varenicline and bupropion sustained-release (SR; combination therapy) compared with varenicline (monotherapy) in cigarette smokers.

DESIGN, SETTING, AND PARTICIPANTS Randomized, blinded, placebo-controlled multicenter clinical trial with a 12-week treatment period and follow-up through week 52 conducted between October 2009 and April 2013 at 3 midwestern clinical research sites.

Five hundred six adult (18 years) cigarette smokers were randomly assigned and 315 (62%) completed the study.

INTERVENTIONS Twelve weeks of varenicline and bupropion SR or varenicline and placebo.

MAIN OUTCOMES AND MEASURES Primary outcome was abstinence rates at week 12, defined as prolonged (no smoking from 2 weeks after the target quit date) abstinence and 7-day point-prevalence (no smoking past 7 days) abstinence.

Secondary outcomes were prolonged and point-prevalence smoking abstinence rates at weeks 26 and 52.

Outcomes were biochemically confirmed.

RESULTS At 12 weeks, 53.0% of the combination therapy group achieved prolonged smoking abstinence and 56.2% achieved 7-day point-prevalence smoking abstinence compared with 43.2% and 48.6% in varenicline monotherapy (odds ratio [OR], 1.49; 95% CI, 1.05-2.12; P = .03 and OR, 1.36; 95% CI, 0.95-1.93; P = .09, respectively).

At 26 weeks, 36.6% of the combination therapy group achieved prolonged and 38.2% achieved 7-day point-prevalence smoking abstinence compared with 27.6% and 31.9% in varenicline monotherapy (OR, 1.52; 95% CI, 1.04-2.22; P = .03 and OR, 1.32; 95% CI, 0.91-1.91; P = .14, respectively).

At 52 weeks, 30.9% of the combination therapy group achieved prolonged and 36.6% achieved 7-day point-prevalence smoking abstinence compared with 24.5% and 29.2% in varenicline monotherapy (OR, 1.39; 95% CI, 0.93-2.07; P = .11 and OR, 1.40; 95% CI, 0.96-2.05; P = .08, respectively).

Participants receiving combination therapy reported more anxiety (7.2% vs 3.1%; P = .04) and depressive symptoms (3.6% vs 0.8%; P = .03).

CONCLUSIONS AND RELEVANCE Among cigarette smokers, combined use of varenicline and bupropion, compared with varenicline alone, increased prolonged abstinence but not 7-day point prevalence at 12 and 26 weeks.

Neither outcome was significantly different at 52 weeks.

Further research is required to determine the role of combination therapy in smoking cessation.

TRIAL REGISTRATION clinicaltrials.gov Identifier: http://clinicaltrials.gov/show/NCT00935818.

Document metadata

Corpus | SciFact

Document ID | 11614737

Evidence checklist

Document 11614737

1. IMPORTANCE Combining pharmacotherapies for tobacco-dependence treatment may increase smoking abstinence.

2. OBJECTIVE To determine efficacy and safety of varenicline and bupropion sustained-release (SR; combination therapy) compared with varenicline (monotherapy) in cigarette smokers.

3. DESIGN, SETTING, AND PARTICIPANTS Randomized, blinded, placebo-controlled multicenter clinical trial with a 12-week treatment period and follow-up through week 52 conducted between October 2009 and April 2013 at 3 midwestern clinical research sites.

4. Five hundred six adult (18 years) cigarette smokers were randomly assigned and 315 (62%) completed the study.

5. INTERVENTIONS Twelve weeks of varenicline and bupropion SR or varenicline and placebo.

6. MAIN OUTCOMES AND MEASURES Primary outcome was abstinence rates at week 12, defined as prolonged (no smoking from 2 weeks after the target quit date) abstinence and 7-day point-prevalence (no smoking past 7 days) abstinence.

7. Secondary outcomes were prolonged and point-prevalence smoking abstinence rates at weeks 26 and 52.

8. Outcomes were biochemically confirmed.

9. RESULTS At 12 weeks, 53.0% of the combination therapy group achieved prolonged smoking abstinence and 56.2% achieved 7-day point-prevalence smoking abstinence compared with 43.2% and 48.6% in varenicline monotherapy (odds ratio [OR], 1.49; 95% CI, 1.05-2.12; P = .03 and OR, 1.36; 95% CI, 0.95-1.93; P = .09, respectively).

10. At 26 weeks, 36.6% of the combination therapy group achieved prolonged and 38.2% achieved 7-day point-prevalence smoking abstinence compared with 27.6% and 31.9% in varenicline monotherapy (OR, 1.52; 95% CI, 1.04-2.22; P = .03 and OR, 1.32; 95% CI, 0.91-1.91; P = .14, respectively).

11. At 52 weeks, 30.9% of the combination therapy group achieved prolonged and 36.6% achieved 7-day point-prevalence smoking abstinence compared with 24.5% and 29.2% in varenicline monotherapy (OR, 1.39; 95% CI, 0.93-2.07; P = .11 and OR, 1.40; 95% CI, 0.96-2.05; P = .08, respectively).

12. Participants receiving combination therapy reported more anxiety (7.2% vs 3.1%; P = .04) and depressive symptoms (3.6% vs 0.8%; P = .03).

13. CONCLUSIONS AND RELEVANCE Among cigarette smokers, combined use of varenicline and bupropion, compared with varenicline alone, increased prolonged abstinence but not 7-day point prevalence at 12 and 26 weeks.

14. Neither outcome was significantly different at 52 weeks.

15. Further research is required to determine the role of combination therapy in smoking cessation.

16. TRIAL REGISTRATION clinicaltrials.gov Identifier: http://clinicaltrials.gov/show/NCT00935818.

Source continuity

The canonical evidence above remains the authoritative retrieval target.